Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 3035
1	User Interface
2	Application Logic-1
3	Application Logic-2
4	Application Logic-3
5	Database
6	Cloud Database
7	File Storage
8	External API-1
9	External API-2
10	Machine Learning Model
11	Infrastructure (Server / Cloud)

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Implement Client Script and UI Policy (Incident)

Reference: ServiceNow Incident Management and Client-side Configuration

Guidelines:

Include all the processes (As an application logic / Technology Block)
Provide infrastructural demarcation (Local / Cloud)
Indicate external interfaces (third party API's etc.)
Indicate Data Storage components / services
Indicate interface to machine learning models (if applicable)

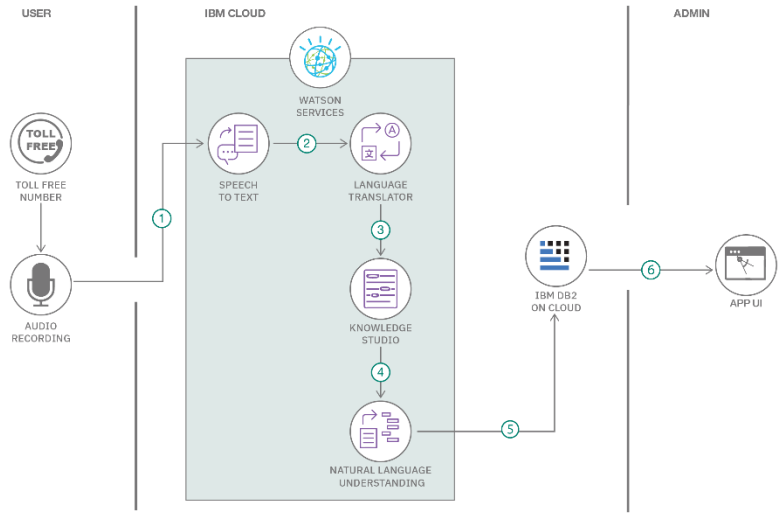


Table-1 : Components & Technologies:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	JavaScript is used for client-side form behavior. ServiceNow scripting follows standard web scripting practices.	JavaScript
2	Security	Role-based access controls, form validation, ACLs, secure scripting practices, and platform authentication are used.	ServiceNow ACL, Roles, Authentication, OWASP practices
3	Scalable Architecture	The solution uses the ServiceNow cloud platform, allowing incident records and client-side rules to operate within the platform architecture.	ServiceNow Platform Architecture
4	Availability	ServiceNow provides a cloud-hosted platform designed for continuous access to incident management functionality.	ServiceNow Cloud Infrastructure
5	Performance	Client-side rules execute only when required, reducing unnecessary server processing and improving incident form responsiveness.	Client Script, UI Policy
1.	Cloud Database	Database Service on Cloud	IBM DB2, IBM Cloudant etc.

2.	File Storage	File storage requirements	IBM Block Storage or Other Storage Service or Local Filesystem
3.	External API-1	Purpose of External API used in the application	IBM Weather API, etc.
4.	External API-2	Purpose of External API used in the application	Aadhar API, etc.
5.	Machine Learning Model	Purpose of Machine Learning Model	Object Recognition Model, etc.
6.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration :	Local, Cloud Foundry, Kubernetes, etc.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	Technology of Opensource framework
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	e.g. SHA-256, Encryptions, IAM Controls, OWASP etc.
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	Technology used
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	Technology used
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	Technology used

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d